

Nathan Chong

EDUCATION

University of Hawai'i at Manoa

Bachelor of Science in Computer Engineering; GPA: 3.83/4.00

Honolulu, HI

Aug 2025 – May 2029

EXPERIENCE

NASA Armstrong Flight Research Center (AFRC)

Aug 2026 – Present

Embedded Systems Intern, Flight Interactive System Technologies

Edwards AFB, CA

- Develop embedded C/C++ and Python software for the MAGEE magnetic capture mechanism and CPSS (Close Proximity Sensor System), integrated onto the Albatross aircraft for autonomous mid-air UAV recapture.
- Integrate RTK GPS (Septentrio mosaic-go) and Doodle Labs mesh radio telemetry with a Cube Orange+ autopilot and Jetson Orin Nano companion computer for real-time relative navigation, then test and validate adaptive RL-based control algorithms for aircraft alignment and recapture through ground and flight testing.

UNLV NSF REU: Smart Cities

May 2026 – Aug 2026

Autonomous Systems Research Intern

Las Vegas, NV

- Adapted a two-stage Rapid Motor Adaptation (RMA) framework, originally built for quadcopters, training a PyTorch policy (PPO, 16M steps) that infers real-time tire grip from 108 LiDAR ranges and motion history alone, deployed on a Traxxas Slash 4x4 with a Jetson and ROS2.
- Closed 44% of the calibration gap to an oracle baseline and reduced cross-grip variance by 20–21% versus a non-adapting policy, validated across 100 simulated episodes per condition on 14 evaluation circuits.
- Confirmed the approach on real hardware across 81 trials and four tire-friction conditions, finding statistically significant changes in steering and yaw-rate behavior ($p = 0.0001$, $p = 0.009$).

Hawaii Space Flight Laboratory (HSFL)

Apr 2026 – Present

Satellite Software Development Engineer Intern

Honolulu, HI

- Configure YAMCS telemetry point-to-address mappings for HiTechSat-1's ground-segment monitoring platform.
- Trace the EPS power-subsystem telemetry pipeline from acquisition through the YAMCS display.

Department of Labor and Industrial Relations

Sep 2024 – May 2025

Robotic Technician Student Intern

Waipahu, HI

- Lead CNC machining operations, designing 20+ custom parts in CAD/CAM and operating Tormach CNC machines for fabrication to resolve high error rates and inconsistent safety practices.
- Mentored a team of 10 on machining safety and best practices, reducing machining errors, accelerating prototyping turnaround, and improving overall safety compliance.

PROJECTS

University Rover Challenge (URC) | GNC Team Member

Aug 2025 – Present

- Integrate YOLO-based object detection and RTK GPS navigation into the rover's autonomy stack, enhancing computer vision and wireless communication systems in Python/C++; contributed to a top-38-of-120+ finish among international teams.

Smart Campus Energy Lab | Electrical Team Member

Aug 2025 – May 2026

- Design, revise, and fabricate PCBs in KiCad; test and validate circuit performance against schematic specifications using oscilloscope-based debugging.

NASA RASC-AL – Project PETAL (Finalist) | Project Manager

Aug 2025 – Jun 2026

- Lead a multi-university team (UH Manoa and UH Hilo), coordinating requirements and integration across energy generation, storage, and distribution subsystems for a scalable lunar power system.
- Selected as one of 14 finalists from 150+ teams nationwide; secured \$7,000 in award funding and presented results to NASA engineers at the RASC-AL Forum.

TECHNICAL SKILLS

Robotics/Embedded: Embedded Systems, Real-Time Flight Software, Sensor Fusion (GPS/Lidar/Vision), Guidance/Navigation/Control, RTK GPS Navigation, ROS2, Jetson

Hardware: PCB Design (KiCad), Oscilloscope Debugging, CNC Machining, CAD/CAM (Onshape, Creo), FDM Printing

Languages: C++, Python, C, MATLAB

Tools: Linux, Docker, Git/GitHub, System Integration, Software Testing & Debugging

AI/ML: PyTorch, Reinforcement Learning (PPO), Convolutional Neural Networks, YOLO, Imitation Learning